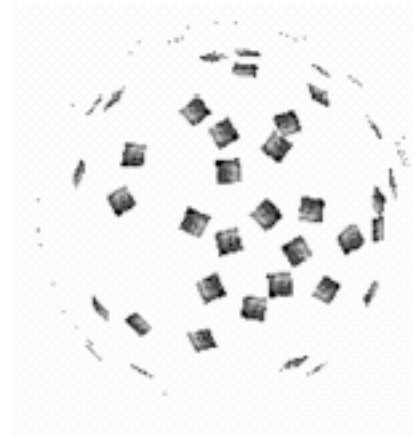
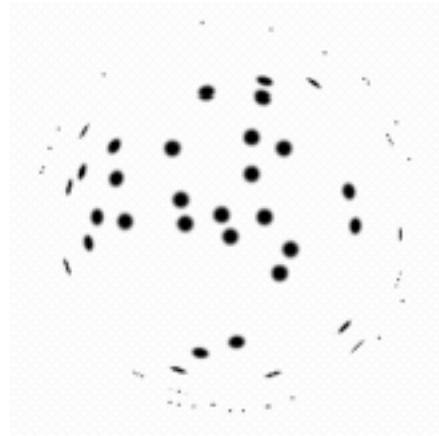
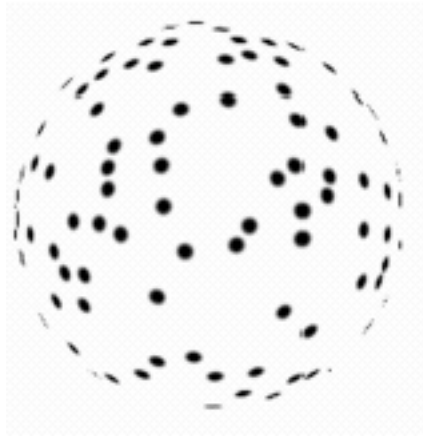


Segmentation and Grouping

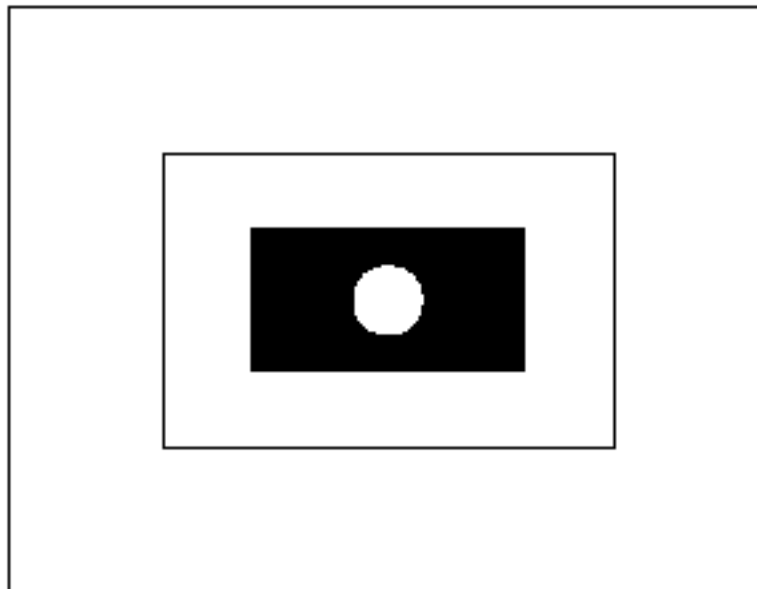
- Motivation: not information is evidence
- Obtain a compact representation from an image/motion sequence/ set of tokens
- Should support application
- Broad theory is absent at present
- Grouping (or clustering)
 - collect together tokens that “belong together”
- Fitting
 - associate a model with tokens
 - issues
 - which model?
 - which token goes to which element?
 - how many elements in the model?

General ideas

- tokens
 - whatever we need to group (pixels, points, surface elements, etc., etc.)
- top down segmentation
 - tokens belong together because they lie on the same object
- bottom up segmentation
 - tokens belong together because they are locally coherent
- These two are not mutually exclusive



Why do these tokens belong together?



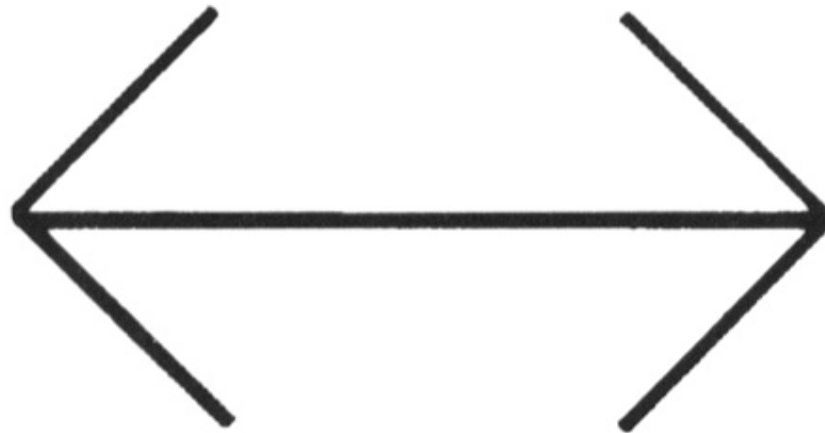
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Set: Segmentation

Slides by D.A. Forsyth

Basic ideas of grouping in humans

- Figure-ground discrimination
 - grouping can be seen in terms of allocating some elements to a figure, some to ground
 - impoverished theory
- Gestalt properties
 - elements in a collection of elements can have properties that result from relationships (Muller-Lyer effect)
 - gestaltqualitat
 - A series of factors affect whether elements should be grouped together
 - Gestalt factors





Not grouped



Proximity



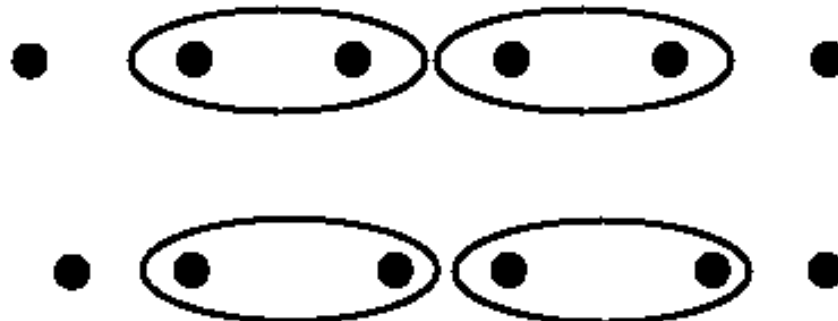
Similarity



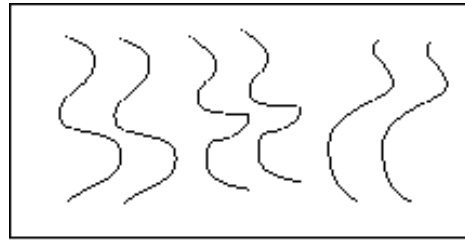
Similarity



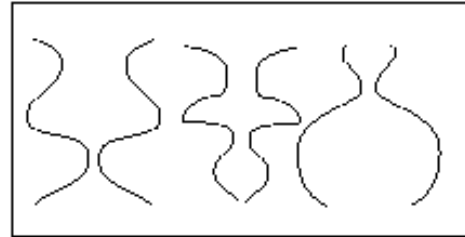
Common Fate



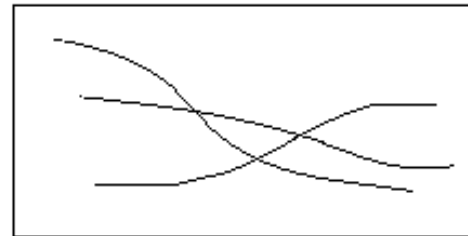
Common Region



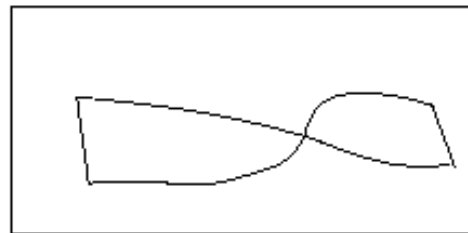
Parallelism



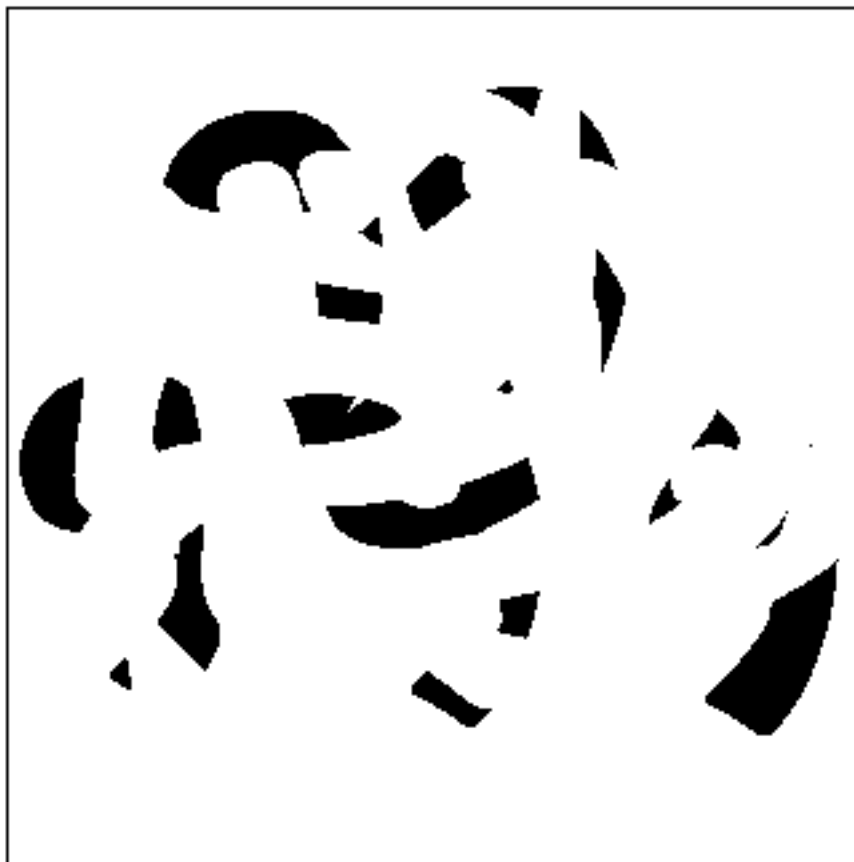
Symmetry



Continuity



Closure



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Set: Segmentation

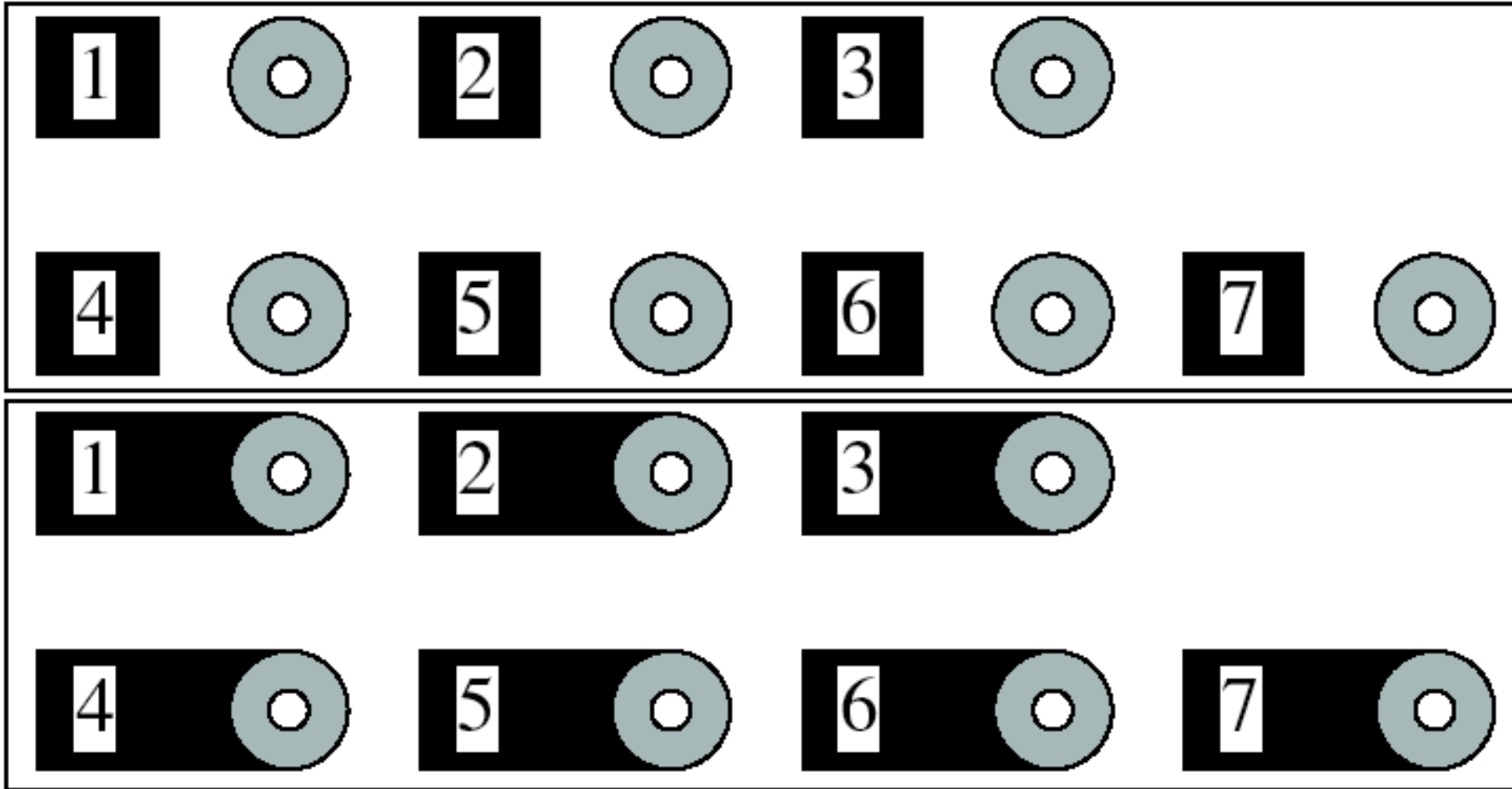
Slides by D.A. Forsyth



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Set: Segmentation

Slides by D.A. Forsyth





Technique: Shot Boundary Detection

- Find the shots in a sequence of video
 - shot boundaries usually result in big differences between succeeding frames
- Strategy:
 - compute interframe distances
 - declare a boundary where these are big
- Possible distances
 - frame differences
 - histogram differences
 - block comparisons
 - edge differences
- Applications:
 - representation for movies, or video sequences
 - find shot boundaries
 - obtain “most representative” frame
 - supports search

Technique: Background Subtraction

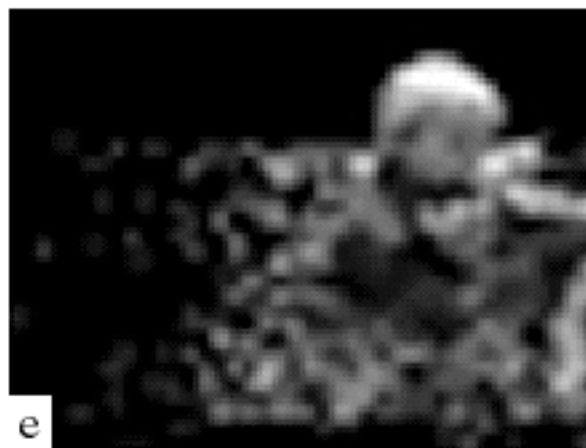
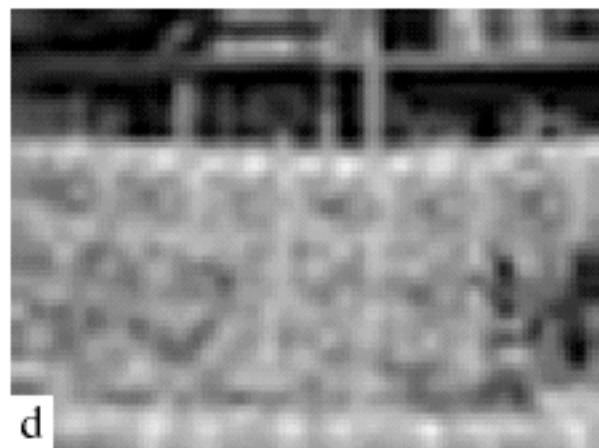
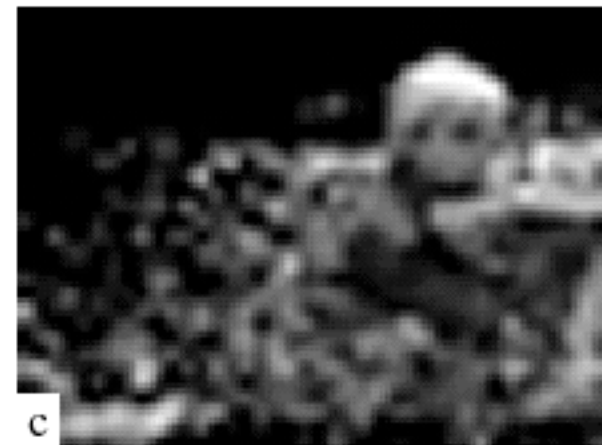
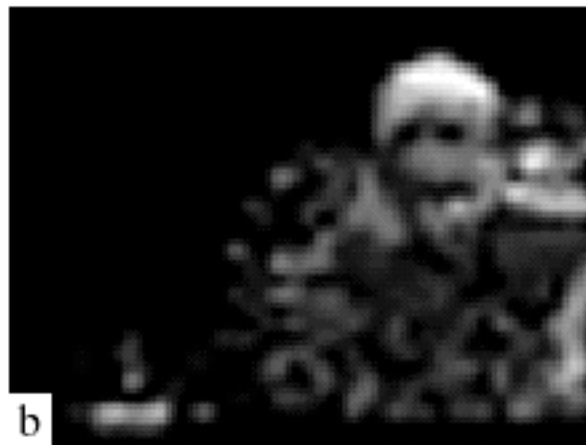
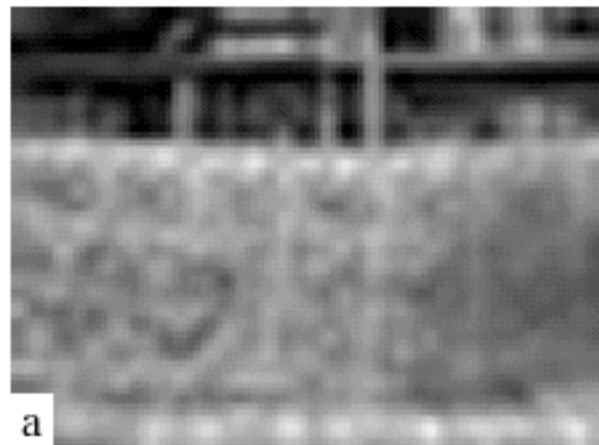
- If we know what the background looks like, it is easy to identify “interesting bits”
- Applications
 - Person in an office
 - Tracking cars on a road
 - surveillance
- Approach:
 - use a moving average to estimate background image
 - subtract from current frame
 - large absolute values are interesting pixels
 - trick: use morphological operations to clean up pixels

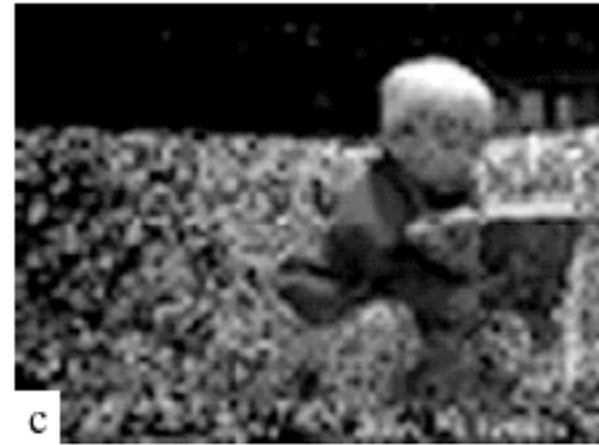
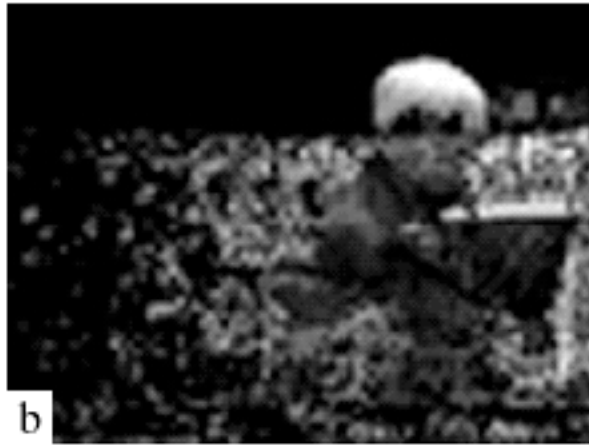


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Set: Segmentation

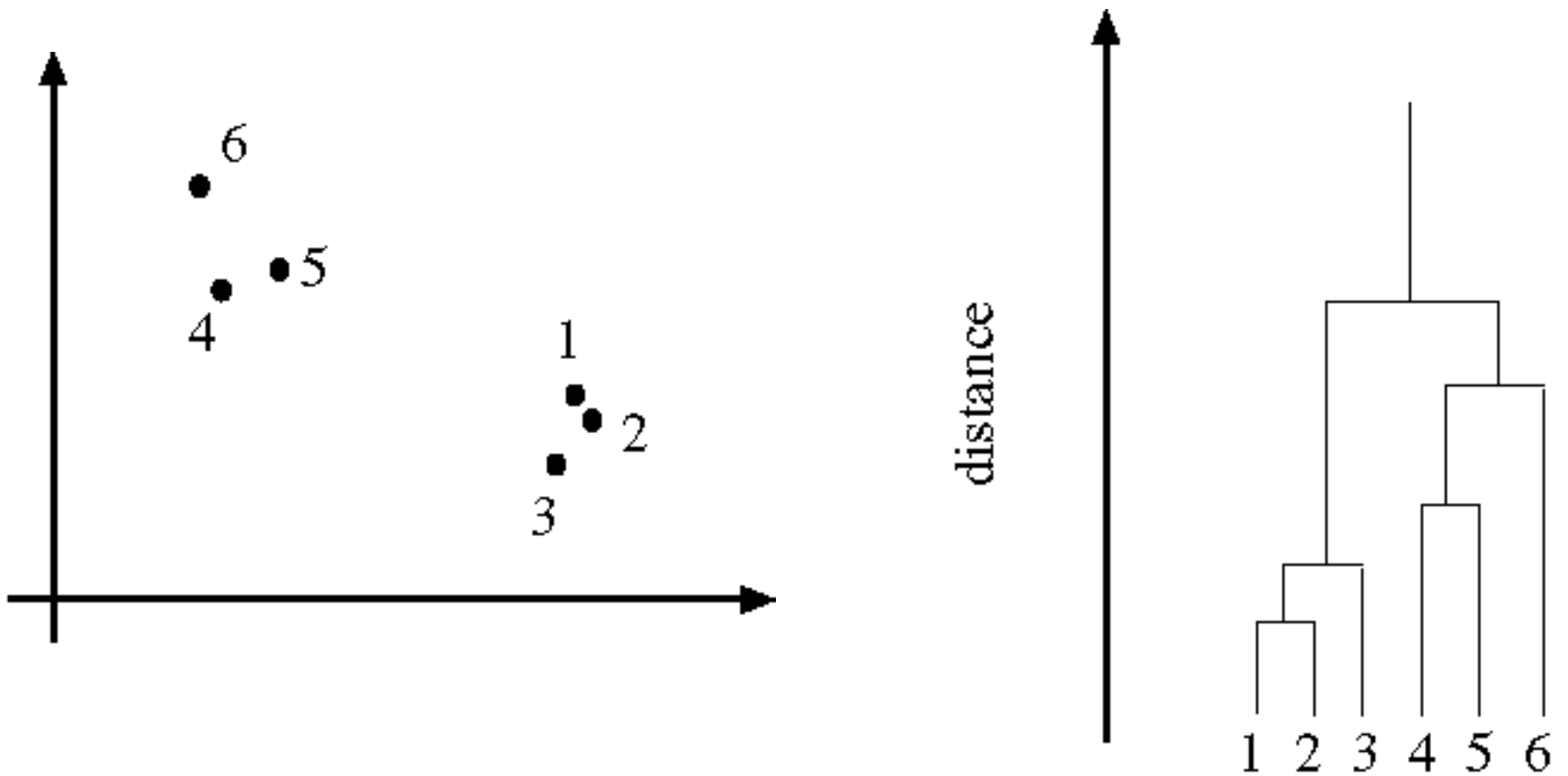
Slides by D.A. Forsyth





Segmentation as clustering

- Cluster together (pixels, tokens, etc.) that belong together
- Agglomerative clustering
 - attach closest to cluster it is closest to
 - repeat
- Divisive clustering
 - split cluster along best boundary
 - repeat
- Point-Cluster distance
 - single-link clustering
 - complete-link clustering
 - group-average clustering
- Dendrograms
 - yield a picture of output as clustering process continues



K-Means

- Choose a fixed number of clusters
- Choose cluster centers and point-cluster allocations to minimize error
- can't do this by search, because there are too many possible allocations.
- Algorithm
 - fix cluster centers; allocate points to closest cluster
 - fix allocation; compute best cluster centers
- x could be any set of features for which we can compute a distance (careful about scaling)

Image



Clusters on intensity



Clusters on color



K-means clustering using intensity alone and color alone



Image

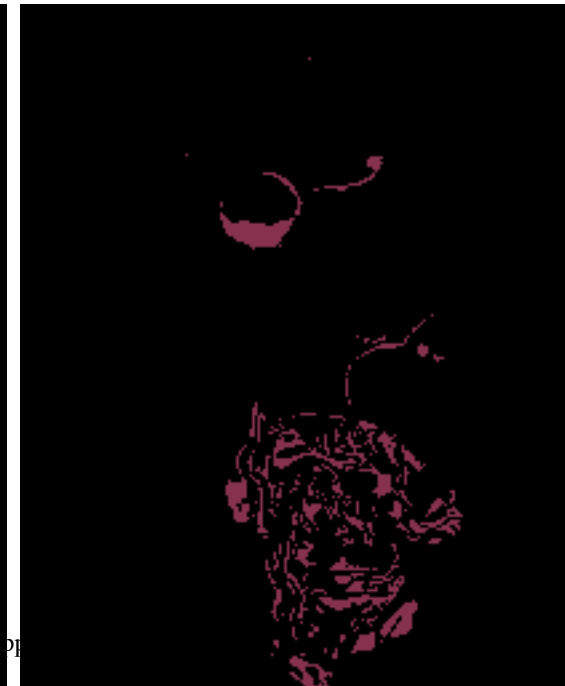


Clusters on color

K-means using color alone, 11 segments



K-means using
color alone,
11 segments.



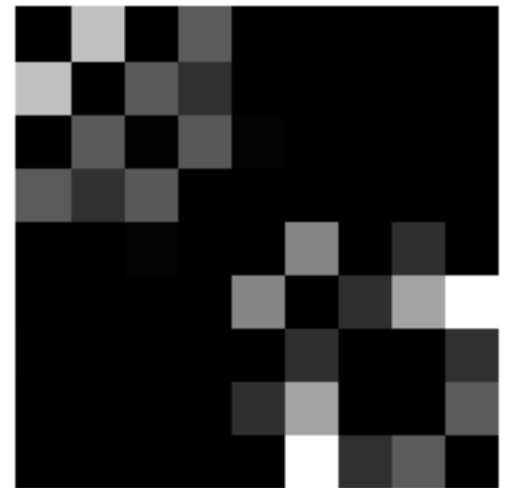
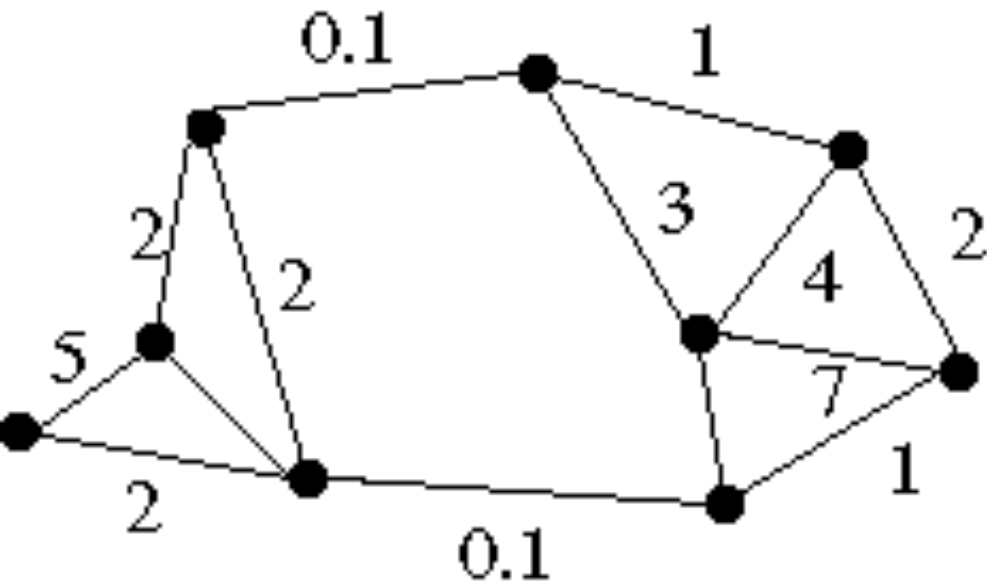


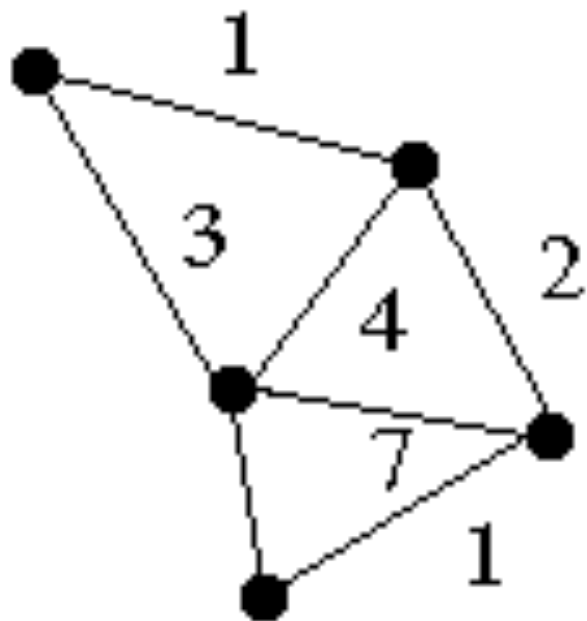
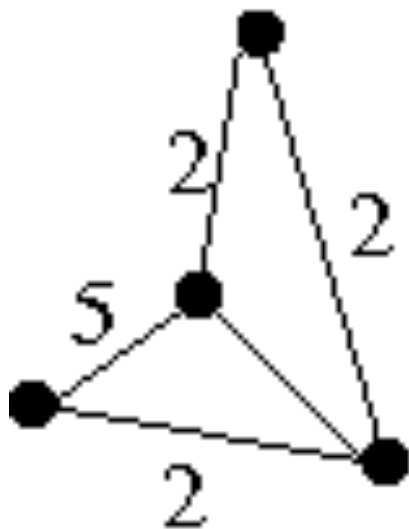
K-means using colour and position, 20 segments



Graph theoretic clustering

- Represent tokens using a weighted graph.
 - affinity matrix
- Cut up this graph to get subgraphs with strong interior links





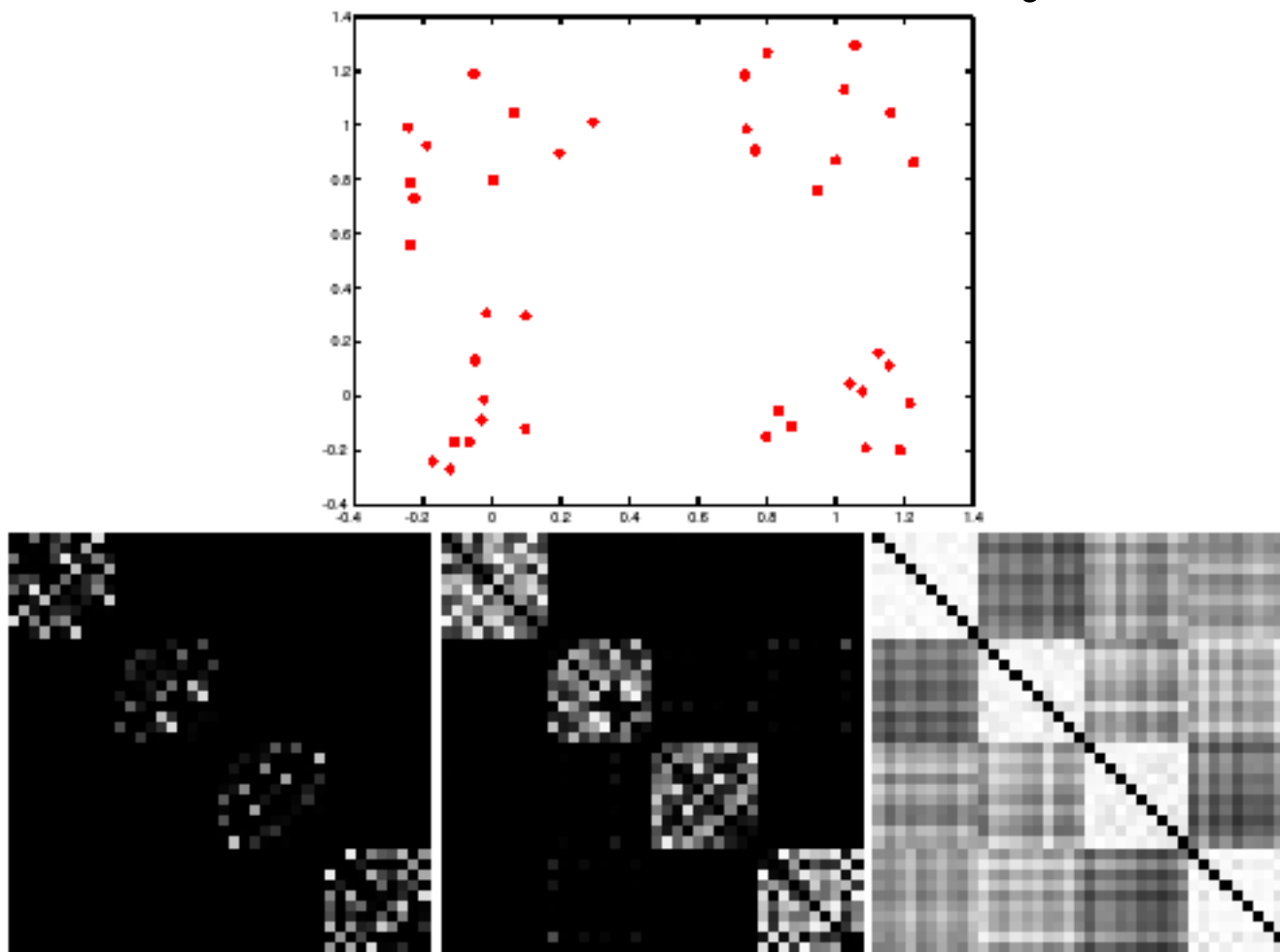
Measuring Affinity

Intensity

Distance

Texture

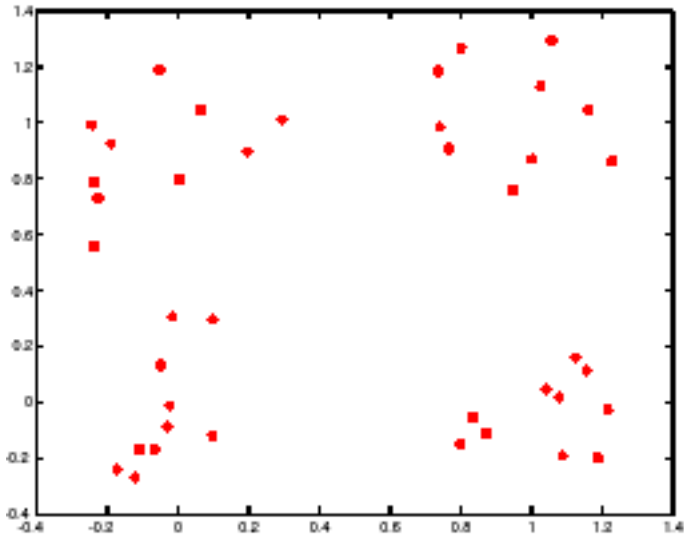
Scale affects affinity



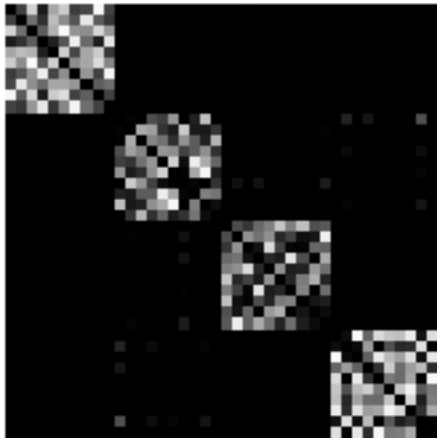
Eigenvectors and cuts

- Simplest idea: we want a vector $\mathbf{v}$ giving the association between each element and a cluster
- We want elements within this cluster to, on the whole, have strong affinity with one another
- We could maximize
- But need the constraint
- This is an eigenvalue problem - choose the eigenvector of A with largest eigenvalue

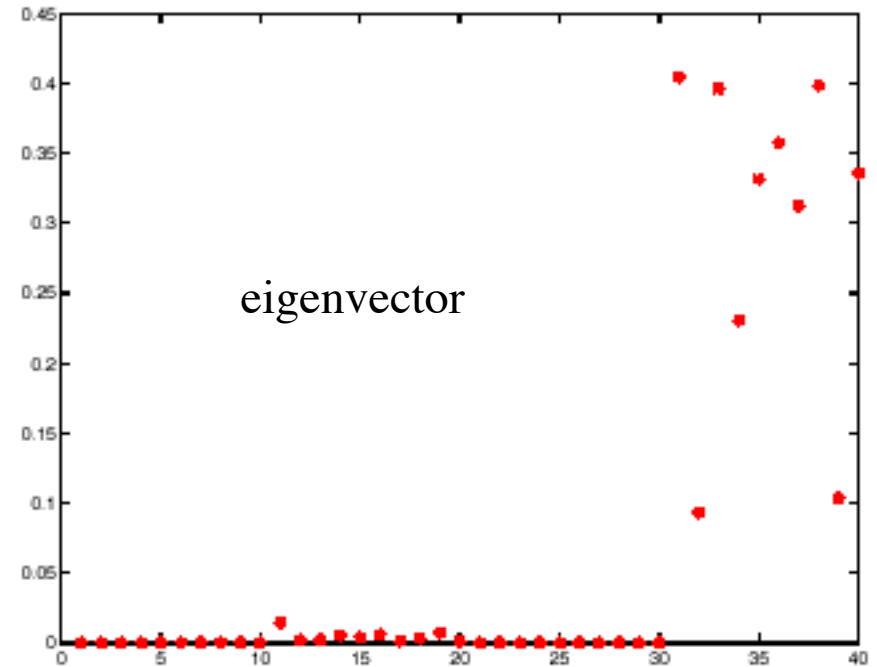
Example eigenvector



points



matrix



eigenvector

More than two segments

- Two options
 - Recursively split each side to get a tree, continuing till the eigenvalues are too small
 - Use the other eigenvectors

Normalized cuts

- Current criterion evaluates within cluster similarity, but not across cluster difference
- Instead, we'd like to maximize the within cluster similarity compared to the across cluster difference
- Write graph as V , one cluster as A and the other as B
- Maximize
- i.e. construct A, B such that their within cluster similarity is high compared to their association with the rest of the graph

Normalized cuts

- Write a vector y whose elements are 1 if item is in A , $-b$ if it's in B
- This is hard to do, because y 's values are quantized
- Write the matrix of the graph as W , and the matrix which has the row sums of W on its diagonal as D , $\mathbf{1}$ is the vector with all ones.
- Criterion becomes
- and we have a constraint

Normalized cuts

- Instead, solve the generalized eigenvalue problem
- which gives
- Now look for a quantization threshold that maximises the criterion ---
i.e all components of y above that threshold go to one, all below go to -b

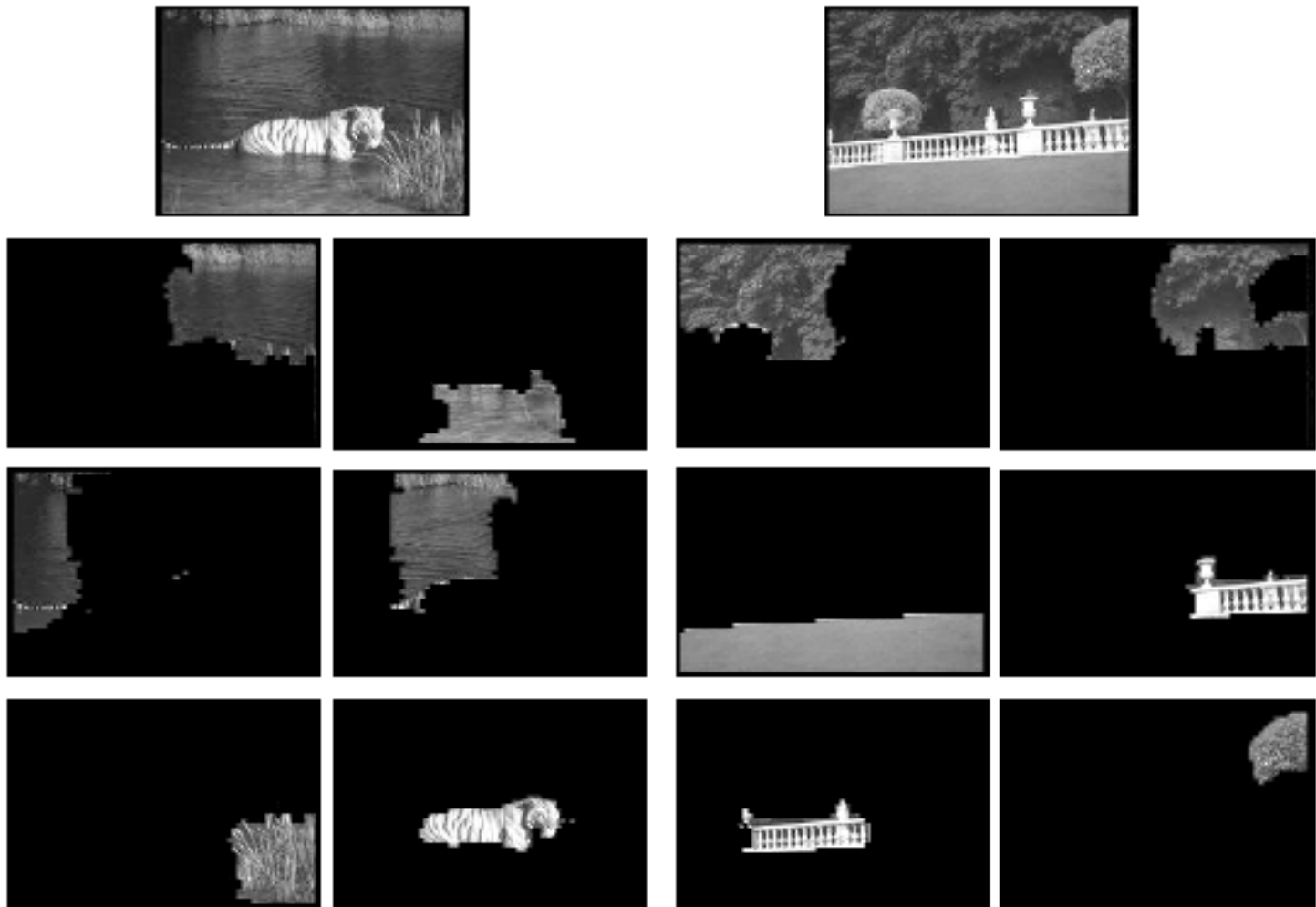


Figure from “Image and video segmentation: the normalised cut framework”,
by Shi and Malik, copyright IEEE, 1998



Figure from “Normalized cuts and image segmentation,” Shi and Malik, copyright IEEE, 2000